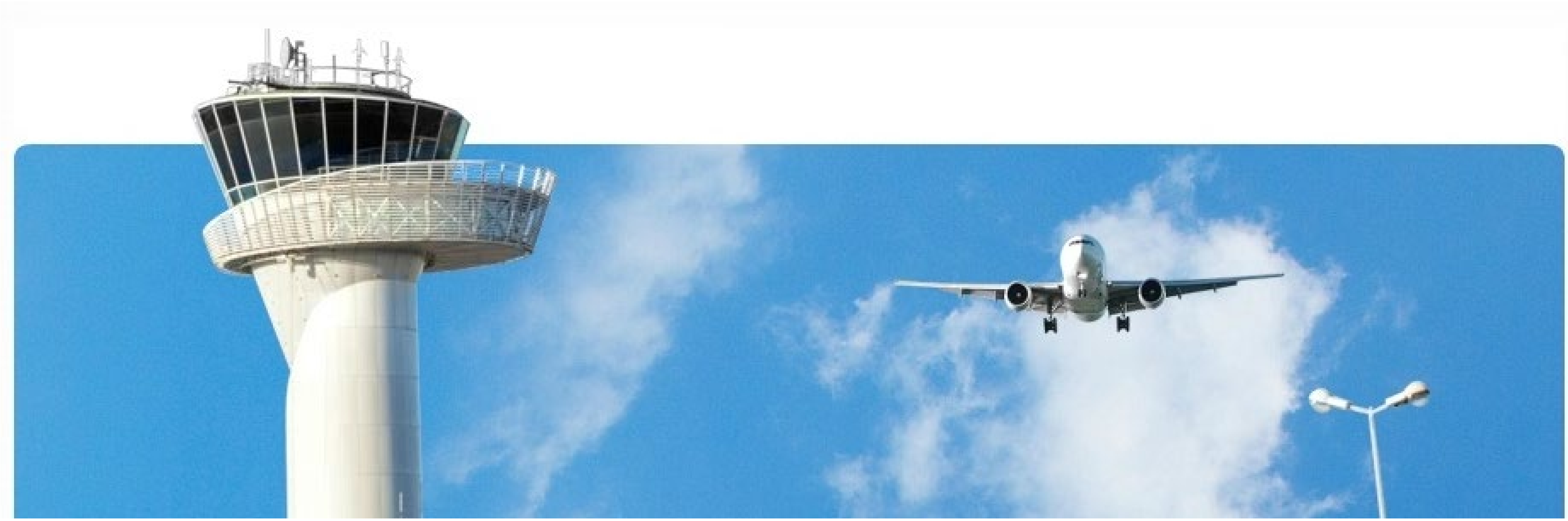
Models for Air Traffic Management



Ground holding problem

08/10/2025

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Introduction

Constrained multi-objective optimization (MOO) problem

Air traffic flow management objective: ensure overall unimpeded flow of aircraft through airspace

→ avoid congestion and delays and, when delays are unavoidable, to reduce as much as possible their impact on airspace users

→ **ground holding problem:** delaying the departure of a flight in order to avoid overloading a capacitated system element

→ **deterministic or stochastic models**

Introduction

In the past, air traffic service providers were the sole decision makers when it came to decisions regarding ground holding.

More recently, however, **Collaborative decision making (CDM)** procedures have led to shared decision making with airspace users.

Example: Airport-Collaborative Decision-Making (A-CDM) helps airports manage better impact of bad weather on their operations by allowing them to disseminate relevant information quickly before disruptions occur + facilitates rapid recovery afterwards

useful for info sharing on de-icing

→ ensure de-iced A/C leave quickly so that they do not ice up again

Introduction

Hypothesis of the GHP (same as for GDP):

• en-route constraints and airport departure constraints: ignored

• only constraint: arrival capacity

→ large multi-airport problem can be decomposed into separate problems, one for each arrival airport

Although control variable = ground delay at the origin airport of each flight, problem can be modeled as one of assigning flights to arrival time intervals at the destination airport

For each flight:

departure time = arrival time - (constant) en-route time

→ amount of ground delay at the origin airport

Model of deterministic GHP

GHP modeled as assigning flights to arrival time intervals at the destination airport while minimizing a cost function.

Define the following inputs: for a set of time intervals t (t = 1, 2, …, T) and set of flights f (f = 1, 2, …, F), let:

b_t = constrained airport's arrival capacity at time interval t,

$ETA(f)$ = scheduled time of arrival: earliest time interval at which flight f can arrive at the constrained airport,

$c_{f\,t}$ = cost of assigning flight f to arrive at time interval t,

and the variable $x_{f\,t}$ = 1 if flight f is assigned to time interval t;

0 otherwise.

[Ball, 2007]

Model of deterministic GHP

Then, deterministic ground holding problem formulated as:

$$\boxed{\min \sum_f \sum_t c_{f\,t} x_{f\,t}}$$

subject to $\sum_f x_{f\,t} \leq b_t$ for all *t*,

$\sum_{t \geq e(f)} x_{f\,t} = 1$ for all *f*,

$x_{f\,t} \geq 0$ and integer for all *f* and *t*.

Suggested definitions for cost function: $c_{f\ t} = r_f\big(t - ETA(f)\big)^{1+\varepsilon}$

or $c_{f\ t} = r_f(t)\big(t - ETA(f)\big)$

→ flight delay costs tend to grow with time at greater than linear rate

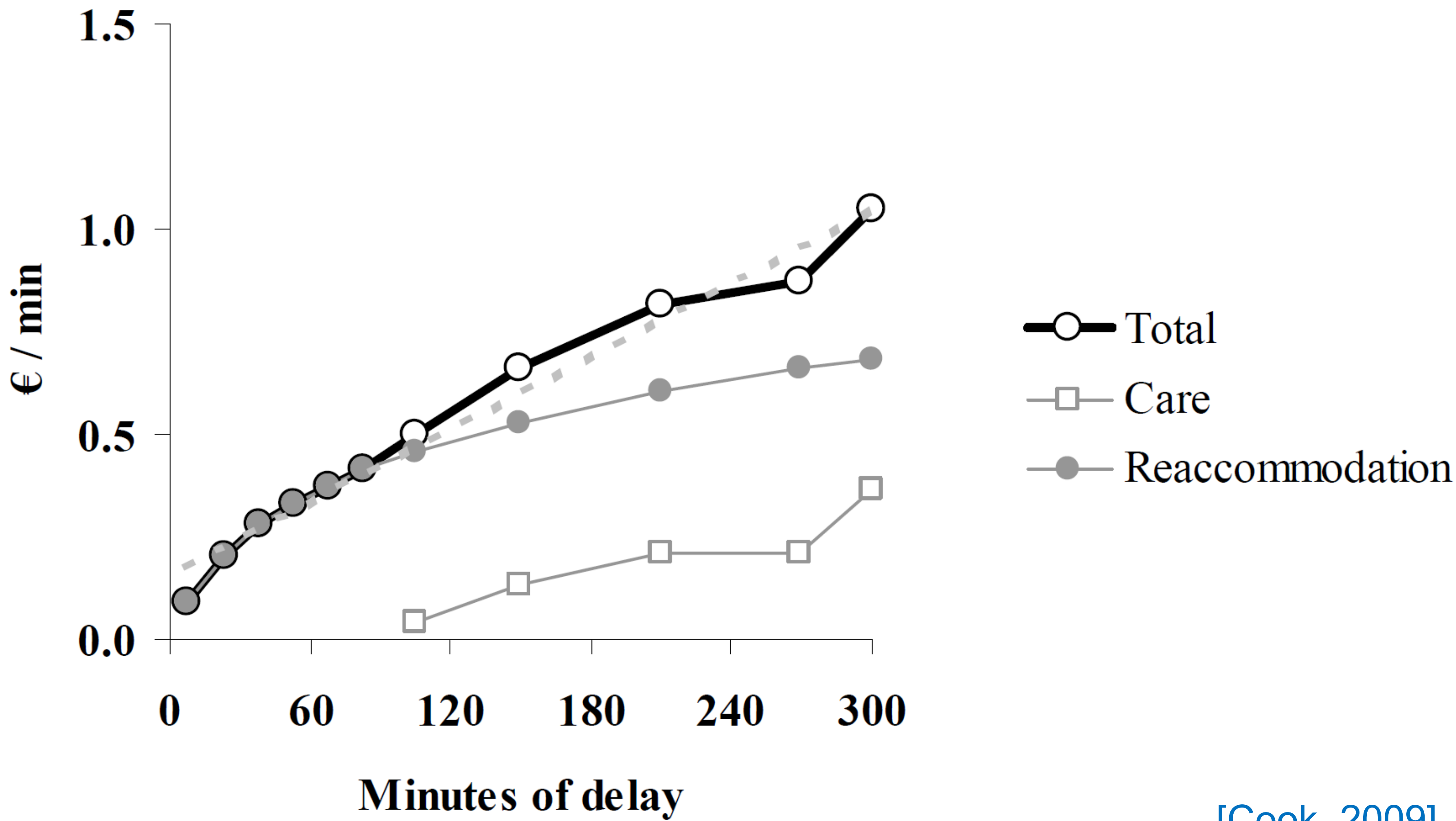
→ Determine r_f for each flight

→ simple transportation model that can be solved very efficiently

[Ball, 2007]

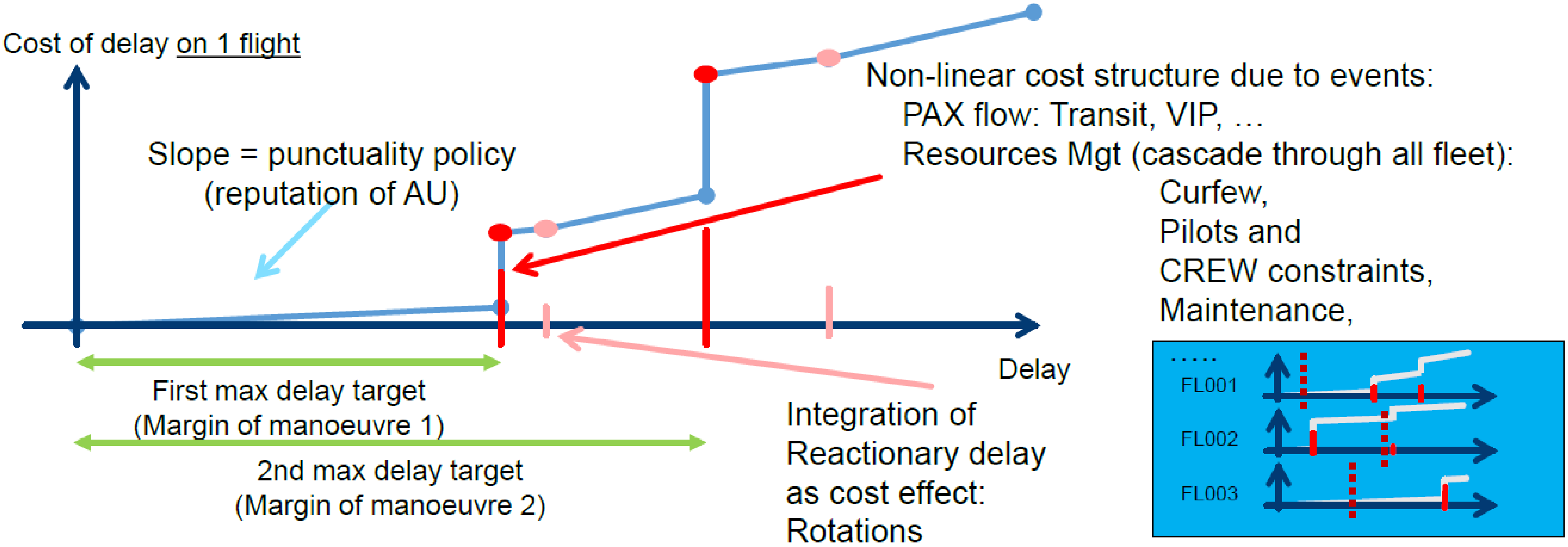
→ **Binary integer programming**

Cost model



[Cook, 2009]

Operational Cost of delay for Airspace Users

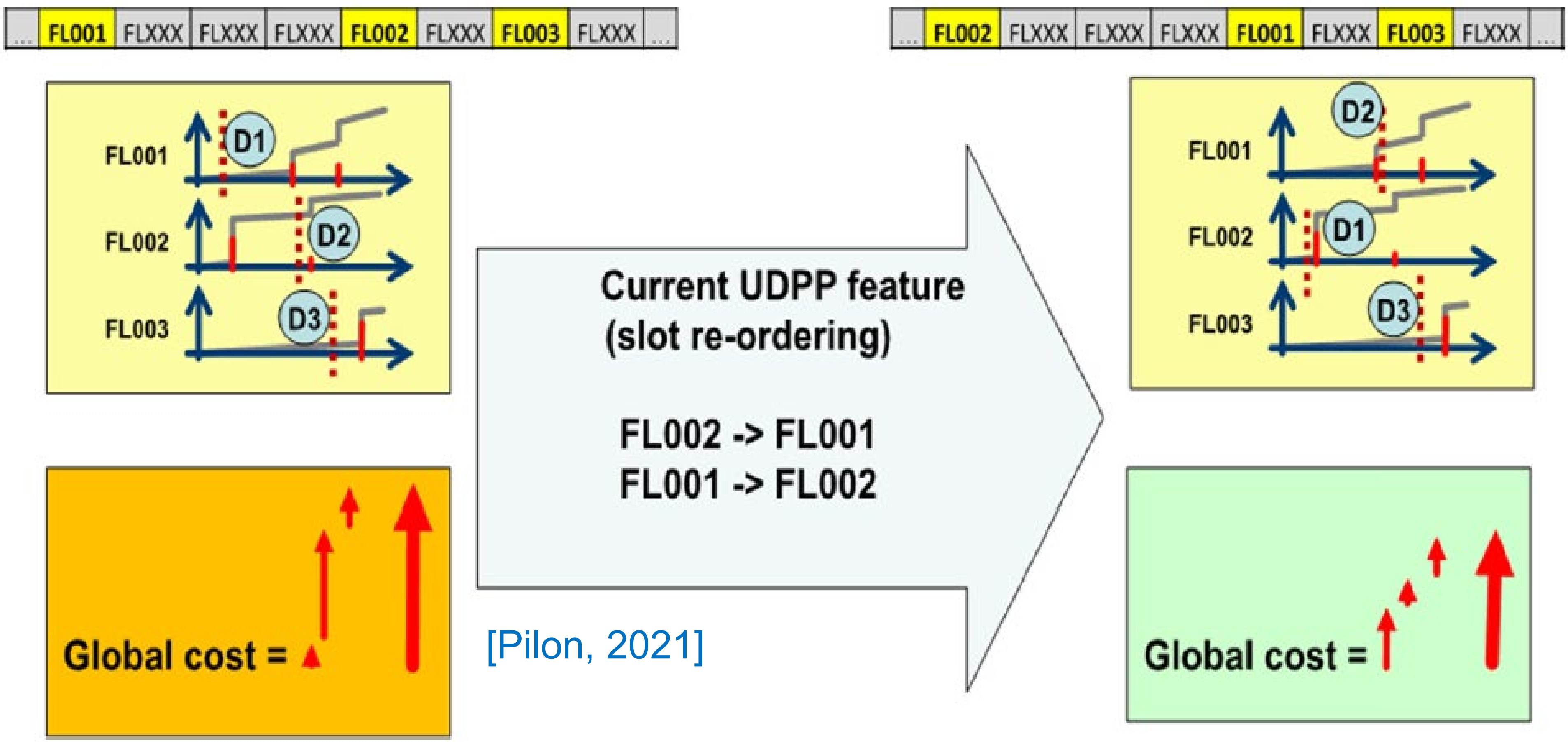


Each flight has its own particular complex cost structure, including fleet rotation impacts, only known by the AU

[Pilon, 2019]

User Driven Priority Process concept

Enables each AU in a capacity constraint to exchange its flight positions (beyond current slot swapping process already implemented by EUROCONTROL Network Manager (NM)) and redistribute its total delay among several of its own flights, reducing cost/impact of delay



[Pilon, 2021]

Further investigations

1. **hub-and-spoke system**: some of passengers from arriving flights A, B and C can be scheduled to transfer at hub to a flight D with a destination common to all of them

→ **banking constraint:** arrivals of flights A, B and C need to be coordinated with departure of D

2. **multi-airport version**: both airport arrival and departure rates are constrained and flights *connected* to each other:

   late arrival of a flight *f* at an airport
   → late departure of 1 or + subsequent flights from that airport

   Because departing flight uses same aircraft as flight *f* or, flight *f* carries many transfer passengers who must board subsequent departing flights

Uncertainty and stochastic models

Uncertainty on multiple levels → failure of many attempts at practical implementation of various air traffic flow management models

Particularly true for models that attempt to optimize over a broad geographic area and/or extended periods of time

→ To be effective, models must either **include stochastic components** explicitly or **address problems restricted to limited geographic and time domains** (for which available information is less subject to uncertainty)

→ **Incorporating uncertainty into ATM planning procedures makes total ATM system more resilient**, because impact of disturbances and propagation of this impact through the system is reduced

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Uncertainty and stochastic models

demand uncertainty = difference between the planned and actual characteristics of the stream of flights arriving at the destination airport.

Due to random or unforeseen events, flights may deviate from their planned departure or arrival times or from their planned trajectories.

This may lead to cancellation or drift (change of planned arrival time) of flights

Examples of factors contributing to demand uncertainty:

- problems in loading passengers onto an aircraft,
- mechanical problems,
- queues on the departure airport’s surface or in the air, and en-route weather problems.

Uncertainty and stochastic models

capacity uncertainty = possibility that random or unforeseen events will cause changes to the maximum achievable flow rates into and out of airports or through airspace waypoints or to the maximum number of flights that can occupy simultaneously a portion of the airspace.

Examples of factors contributing to capacity uncertainty:

- include weather conditions at an airport or in the en-route airspace,
- failures or degradation of Air Traffic Control equipment,
- unavailability of ATC personnel + random changes in flight sequences,
- aircraft mix that requires alterations of anticipated flight departure or arrival spacing.

Uncertainty and stochastic models

Research focused on ground holding models that **explicitly take into account uncertainty in airport arrival capacity**.

Performance of a GDP/GHP planning and control system can be evaluated based on 3 measures:

- total assigned ground delay,
- total airborne delay,
- utilization of the available arrival capacity.

Uncertainty and stochastic models

Planning and control of a GDP/GHP: must balance possibility of:

- too low arrival rate (→ under-utilization of the server)

- with too high arrival rate (→ large airborne queue and excessive airborne delay)

If it were known that such a scenario would occur with certainty

→ a deterministic ground holding model could be used with this scenario providing capacity constraints input

By contrast, stochastic models treat AAR as a random variable and use as input several such scenarios together with associated probabilities

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Ground holding models with stochastic airport capacity

Because of demand and capacity uncertainty, the *best* policy calls for some degree of airborne delay in order to ensure an acceptably intensive utilization of the available arrival capacity

Starting point for defining airport arrival capacity: specifying number of flights that can land within a time interval (AAR), and characterize AAR over a period of time (e.g. 6h)

Typical GDP/GHP caused by weather disturbance moving through a region:

1. AAR would start at its normal level (e.g. 60 arrivals per hour),
2. AAR would decrease to 1 or more degraded levels (e.g. 30 arrivals per hours), for several time intervals (e.g. 4 hours),
3. AAR would then return to its original level.

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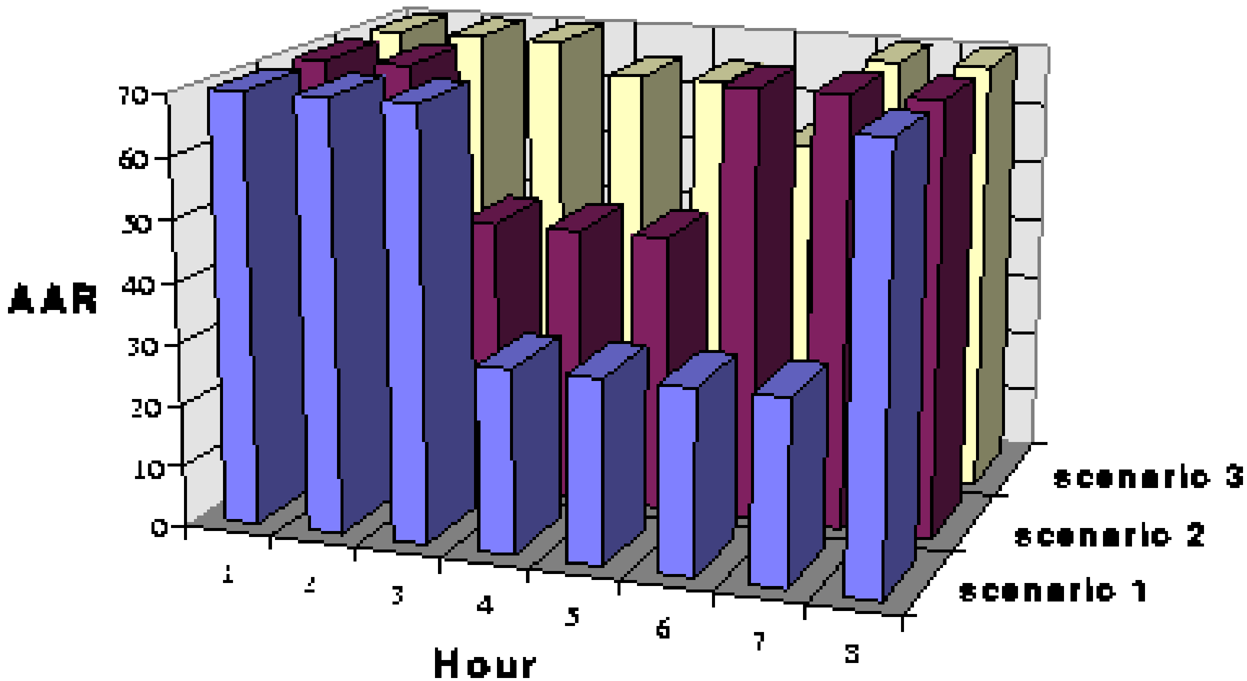
Ground holding models with stochastic airport capacity

Ex: possible multiple AAR scenario forecast for a fictitious airport whose normal AAR is 70 flights per hour:

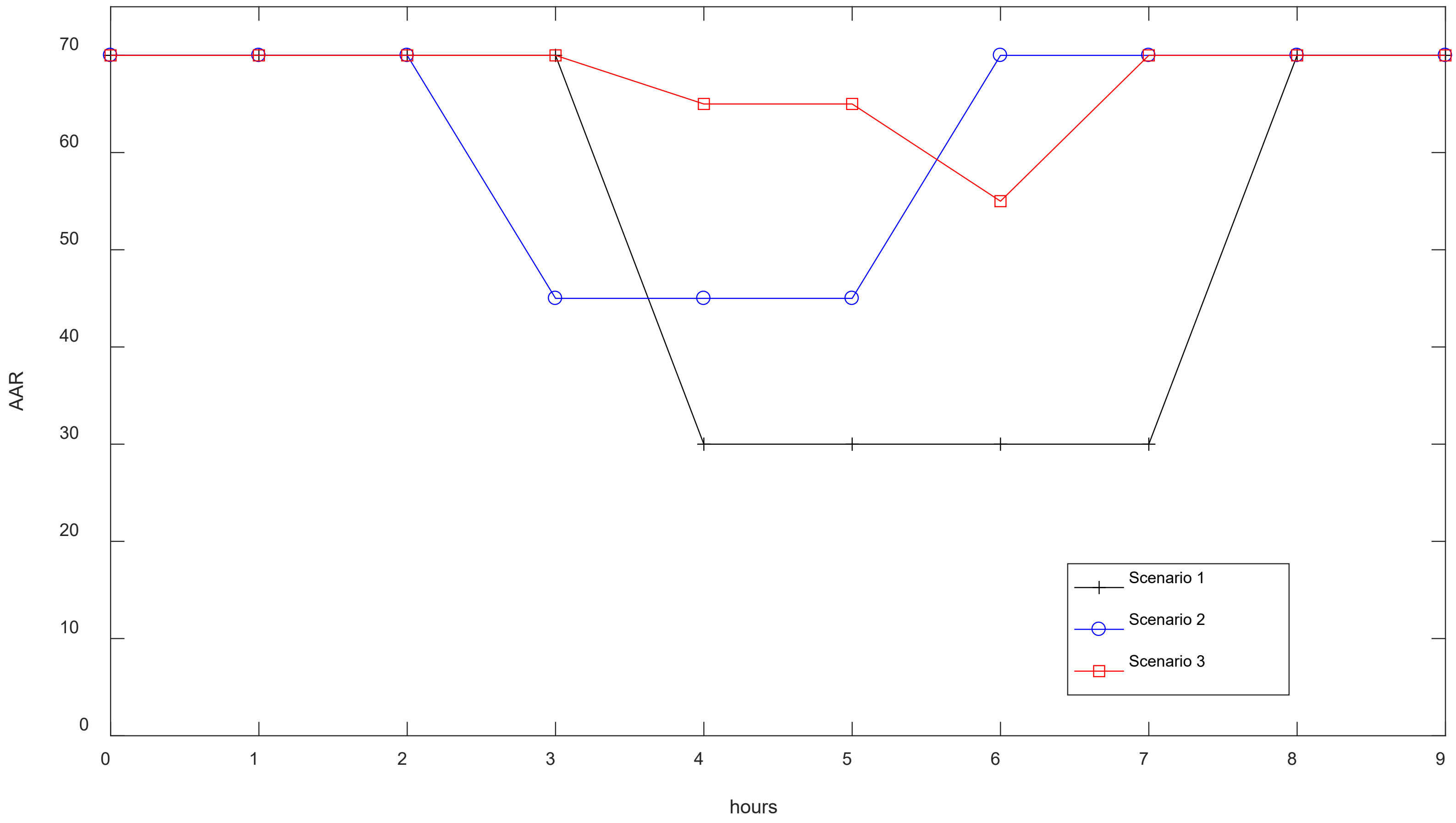


But there could be greater number of scenarios to consider

→ greatly obscures the GDP that minimizes overall delay costs and renders almost useless any simple, intuitive approach for finding it

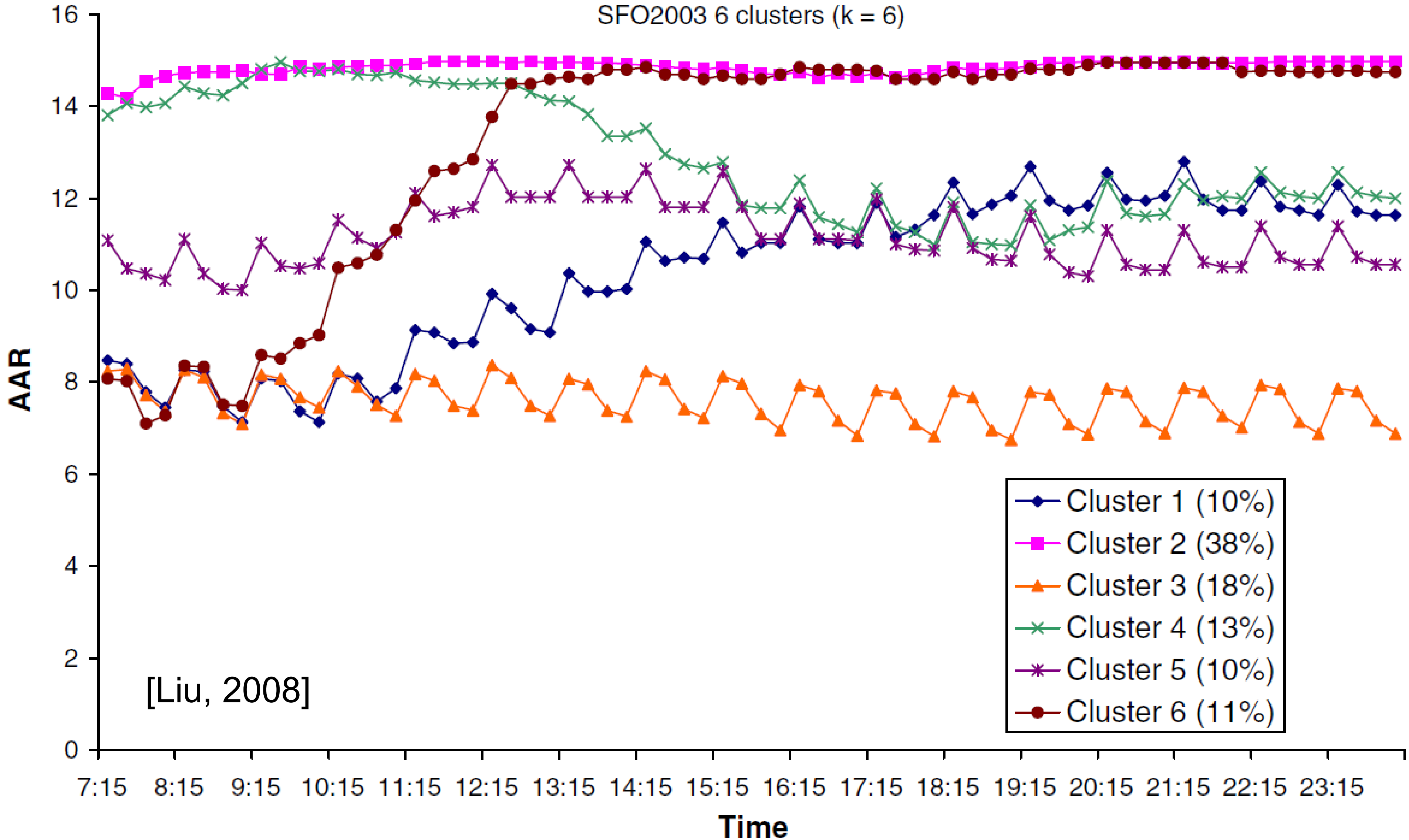
Ground holding models with stochastic airport capacity

Ex: possible multiple AAR scenario forecast



Ground holding models with stochastic airport capacity

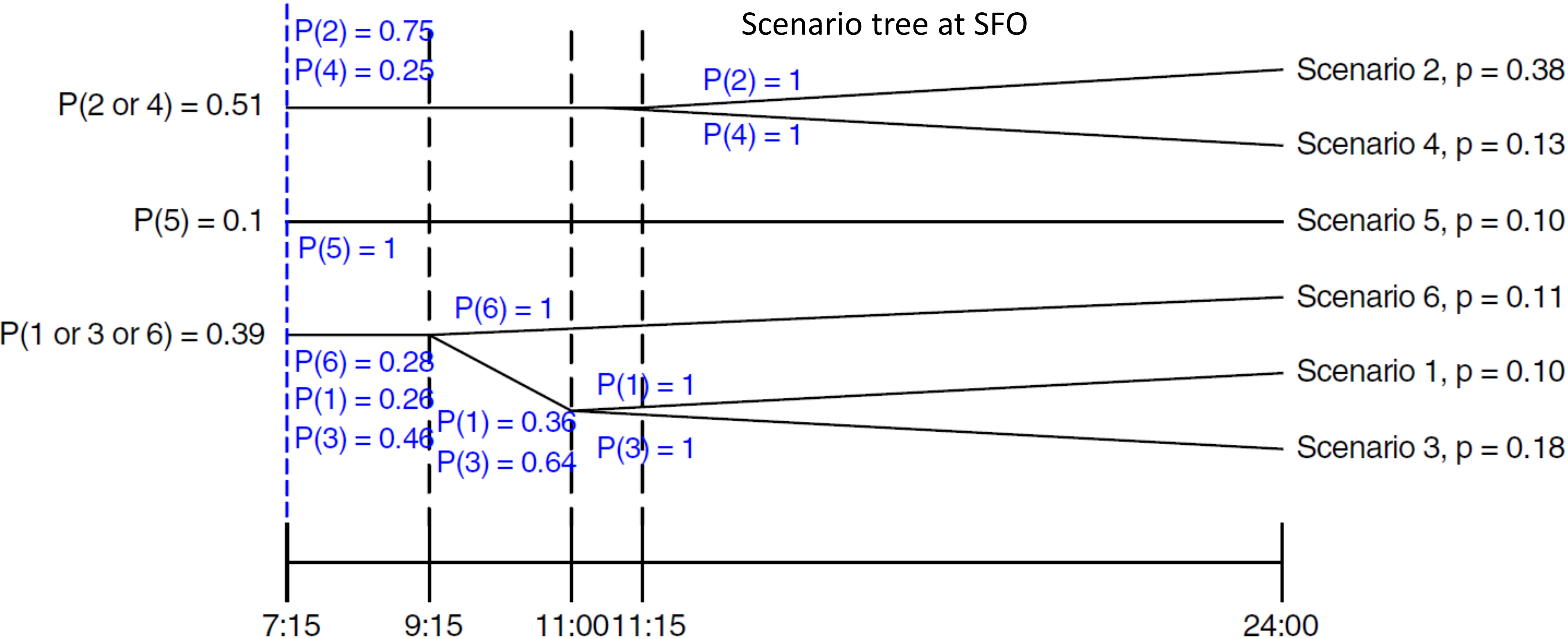
AAR scenarios with probabilities from the *clusters* (patterns/groups of arrival capacity profiles): Ex. of scenarios at SFO:



Ground holding models with stochastic airport capacity

AAR scenario tree with probabilities from the clusters:

→ Application to *dynamic* stochastic models



scenario tree: set of capacity scenarios with associated probabilities of realization that is organized into a tree where a *branching point* represents the time when initially similar scenarios evolve into distinct scenarios

[Liu, 2008]

Ground holding models with stochastic airport capacity

Objective function for the *static* model:

Minimizes: cost of ground delay + expected cost of airborne delay

With c^a, cost of holding one flight in the air for one time period, objective function is defined as:

p_q = probability of the occurrence of scenario *q* for *q* = 1,…, *Q*

$$\boxed{\min\left(\sum_f \sum_t c_{ft} x_{ft} + \sum_q p_q \sum_t c^a y_{t-1q}\right)}$$

y_{tq} = number of flights held in the air from time period *t* to *t* + 1 under scenario *q* for *q* = 1,…, *Q*

cost of ground delay

expected cost of airborne delay

Ground holding models with stochastic airport capacity

Dynamic models use evolving information (ex: branching point that occurs at a particular time of day) about capacity to make ground holding decisions

When a branching point is reached → new information about which of the branches is being realized on a particular day

Based on this information, ground-hold decisions for flights that have not yet taken off can be made or revised

For example, a flight scheduled to depart just before a branching point might be held so that the release decision can be made with the benefit of the information that will soon become available

Exercises: programming GHP with Matlab

Simple model of deterministic GHP: “toy problem”

Consider 3 arrival time slots: t = {[1600, 1601]; [1602, 1603], [1604, 1605]}

3 constrained airport’s arrival capacities: $b_1 = 1$, $b_2 = 2$, $b_3 = 1$

4 aircrafts, whose scheduled times of arrival are:

ETA(1) = 1600, ETA(2) = 1600, ETA(3) = 1602, ETA(4) = 1603

and whose cost parameters are:

$r_1 = 1$, $r_2 = 3$, $r_3 = 1$, $r_4 = 5$

We consider here a linear cost function: $c_{f\,t} = r_f\left(t - ETA(f)\right)$

The problem to solve is: $$\min \sum_{f=1}^{4}\sum_{t=1}^{3} c_{f\,t}\,x_{f\,t}$$

+ apply corresponding constraints

Simple model of deterministic GHP: "toy problem"

Integer linear programming, define variables:

x_1	x_2	x_3	sum = 1
x_4	x_5	x_6	sum = 1
x_7	x_8	x_9	sum = 1
x_{10}	x_{11}	x_{12}	sum = 1
sum ≤1	sum ≤2	sum ≤1	

+ constraint: "*aircraft can not arrive before earliest due time*"

or:

x_1	x_2	x_3	sum = 1
x_4	x_5	x_6	sum = 1
	x_7	x_8	sum = 1
	x_9	x_{10}	sum = 1
sum ≤1	sum ≤2	sum ≤1	

Use for example `intlinprog` function of Matlab to solve this problem:

1. Define the equality and inequality matrices corresponding to the equality and inequality constraints (that will be the arguments of the function `intlinprog`).
2. Calculate each aircraft's arrival time and the total cost obtained with a GHP formulation.

MGTA references

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- [Pilon, 2021] N. Pilon et al., User-Driven Prioritisation Process (UDPP) from advanced experimental to pre-operational validation environment, Journal of Air Transport Management 97, 102124, 2021 doi.org/10.1016/j.jairtraman.2021.102124